



A Holistic Approach to Sustainable, Digital EU Agriculture, Forestry, Livestock and Rural Development based on Reconfigurable Aerial Enablers and Edge Artificial Intelligence-on-Demand Systems

CHAMELEON Bundle User Manual. Crop water stress due to drought

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GLOSSARY OF TERMS

Term	Description

LIST OF ABBREVIATIONS AND ACRONYMS

Abbreviation	Meaning
CWSI	Crop Water Stress Index
Tc	Canopy Temperature
Ta	Ambient temperature
RH	Relative humidity

1 EXECUTIVE SUMMARY

EXEC SUM

This document describes how to calculate the Crop Water Stress Index (CWSI) using thermal orthomosaic data and climatic data (air temperature and relative humidity) to monitor vine water stress. The main advance compared with existing procedures is that it calculate the index for each individual plant. The user must configure relevant parameters, including air temperature, relative humidity, and date extraction, and select the appropriate input files such as the vines frames shapefile and thermal orthomosaic. It can generate thematic maps of stress and also generate alarms choosing the preferable segmentation method (k-means or percentile). The output, containing CWSI and temperature values of well illuminated vegetation, can then be loaded into QGIS or any WEBGIS viewer for detailed visualization and analysis, aiding in the assessment of crop water stress.

2 INTRODUCTION

2.1 PURPOSE AND SCOPE

CROP WATER STRESS INDEX USING THERMAL ORTHOMOSAIC

The main objective is to calculate and analyse the Crop Water Stress Index (CWSI) for each individual vine using thermal orthomosaic data. This involves configuring parameters, selecting appropriate input files, and using segmentation methods to generate detailed CWSI and temperature values. The final output is CWSI index raster that can be visualized in QGIS or any WEBGIS viewer.

The CWSI is based on the determination of the canopy temperature and its difference with air temperature ($T_c - T_a$). Also, it requires the calculation of the vapor pressure deficit from air temperature and relative humidity. The threshold of stressed and non-stressed vines is defined by the upper baseline and lower baseline, respectively. These baselines are specific for each species and even for each variety of the same species. To feed the tool, a literature review was performed to determine the most appropriate values for vineyard, but it can be accommodated by the user based on their own experience. The CWSI is calculated for each individual vine in the vineyard. Thus, thematic maps and alarm generation can be performed.

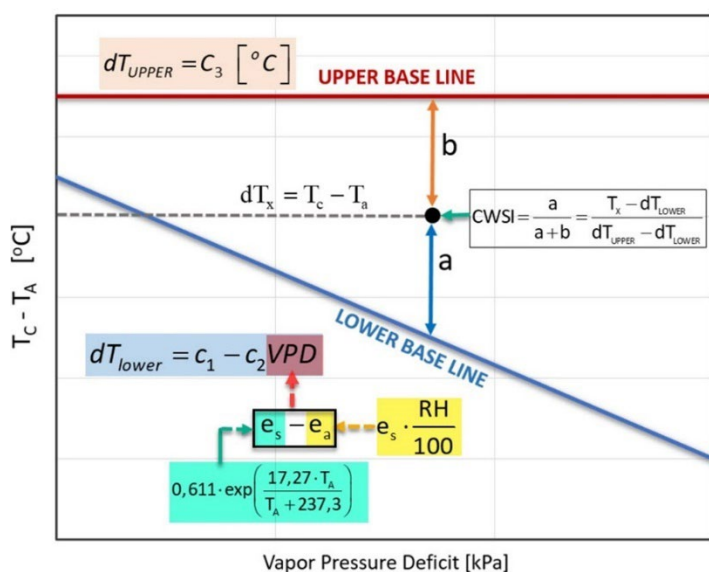


Figure 1. Graphical representation of the determination of CWSI

2.2 APPROACH

2.2.1 USER INTERFACE OF PAFYC TOOLS AND MAIN PARAMETERS

The user should select Crop Water Stress Index using Thermal orthomosaic in the tool (Fig. 1).

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Figure 2. PAFyC tools gui with the bundle selected.

After select the parameters for this command, the following parameters should be defined (Fig. 2.)

Parameters manager for command: Crop Water Stress Index using Thermal Orthomosaic

	Code	Enabled	Value	Description
1	CWSITHO_AirTemperature	Yes	33.1	Air temperature (Celsius degrees)
2	CWSITHO_Date	Yes		A valid date in selected format or empty when read from tag in thermal orthomosaic file name
3	CWSITHO_DateFormat	Yes	%d/%m/%Y	A valid date format for datetime.strptime function
4	CWSITHO_DateFromThermalOrthomosaicFile	Yes	True	Read date from tag in thermal orthomosaic file name
5	CWSITHO_FactorToTemperature	Yes	0.010000	Multiplicative factor to convert raster values in thermal orthomosaic to temperature
6	CWSITHO_FramesShapeFile	Yes	E:/PAFyCToolsGui/20230825_Tarazona_Thermal_CWSI/input/Shape_cepas.shp	Plants frames shapefile (SHP)
7	CWSITHO_LowerBaselineCoefficientA	Yes	-2.016800	Lower base line coefficient a, line slope (adimensional)
8	CWSITHO_LowerBaselineCoefficientB	Yes	2.702	Lower base line coefficient b, value of y when x=0 (celsius degree)
9	CWSITHO_MaximumValueForPercentile	Yes	0.20	Maximum value for percentile segmentation (per unit)
10	CWSITHO_NumberOfClustersForKmeans	Yes	3	Number of clusters for kmeans, lower is canopy
11	CWSITHO_OutputShapeFile	Yes	E:/PAFyCToolsGui/20230825_Tarazona_Thermal_CWSI/output/20240524/shapes_cepas_cwsis.shp	Output save file or empty for use input plants frames shapefile
12	CWSITHO_RelativeHumidity	Yes	28.7	Relative humidity (percentage)
13	CWSITHO_SegmentationMethod	Yes	kmeans	Segmentation method: kmeans or percentile
14	CWSITHO_ThermalOrthomosaicFile	Yes	E:/PAFyCToolsGui/20230825_Tarazona_Thermal_CWSI/input/ORT_FDA_Tarazona_P001_20230825_4258_5782_66_mmm.tif	Thermal orthomosaic file (GeoTIFF)
15	CWSITHO_ThermalOrthomosaicFileTagDatePosition	Yes	5	Thermal orthomosaic file tag date position, starts in 1
16	CWSITHO_ThermalOrthomosaicFileTagsStringSeparator	Yes	-	Thermal orthomosaic file tags string separator

Figure 3. List of parameters to be selected in the tool

The user should configure the necessary parameters for the command. This includes selecting and adjusting all relevant parameters. Also, the user should ensure that the air temperature is set correctly. It is crucial to read the date from the orthomosaic, especially if an orthomosaic file is being used as input data. The date must be in the correct position between the defined tag ("_" in this case) within the file name, so that the tool can identify it correctly. Additionally, if the temperature is multiplied by 100 in the raster layer, convert it to degrees Celsius by dividing by 100 (FactortoTemperature parameter set to 0.01). Also, the user should select the

input shapefile which contains the definition of the vine frames, and the raster orthomosaic which contains the temperatures.

After that and with the aim to perform vegetation segmentation, the user should select the number of clusters for kmeans, by setting the value to 3 (calibrated with field data). The cluster with the lower temperature value will be considered vegetation (soil temperature is always values of each plant).



Figure 4. Example of canopy segmentation using k-means and temperature information.

Next, the user should configure the output shapefile. By selecting the output file path and ensuring that the output file name is correctly placed in the designated path.

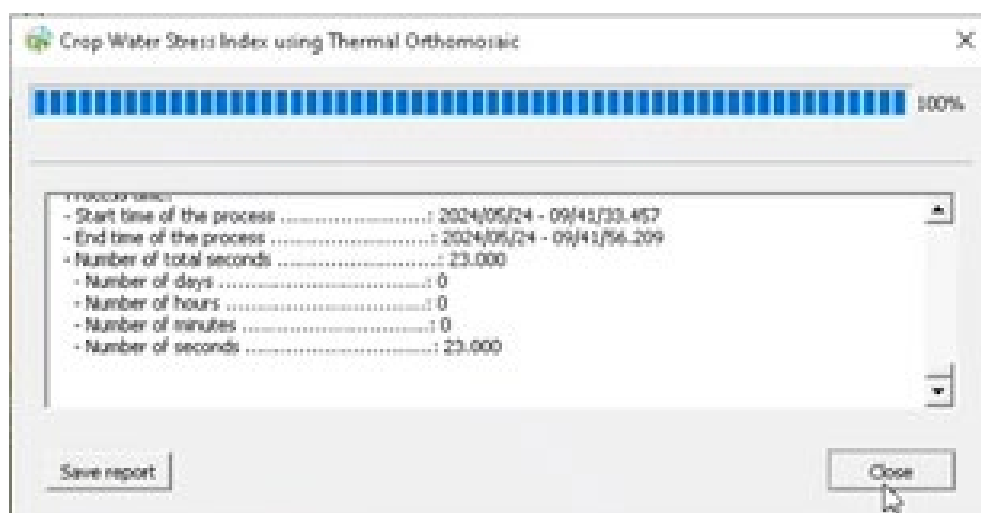
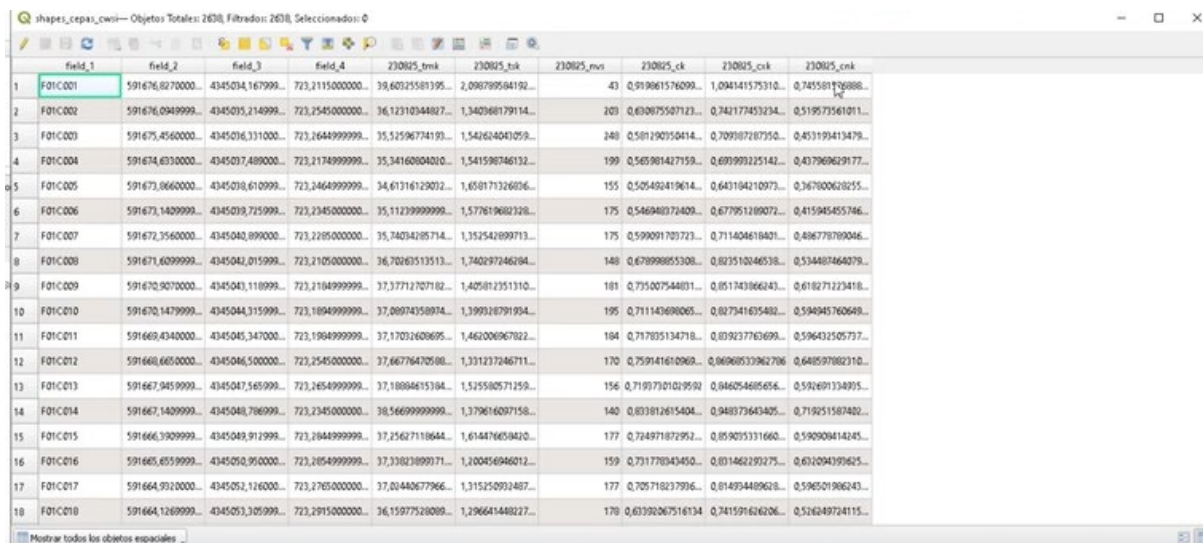


Figure 5. Report of the execution

The output file contains calculated values of the Crop Water Stress Index CWSI (mean(ck), maximum (cxk), and minimum (cnk)) and temperature (mean (tmk), standard deviation,(tsk), and the number of pixels segmented as vegetation (nvs)).

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	field_1	field_2	field_3	field_4	230025_bmk	230025_mv	230025_ck	230025_cnk
1	F01C001	591676,0270000...	4345034,167999...	723,2115000000...	39,60325581395...	2,098799504192...	43	0,919661576099...
2	F01C002	591676,0949999...	4345035,214999...	723,2545000000...	36,12310344887...	1,340268179114...	203	0,630875507123...
3	F01C003	591675,4560000...	4345036,331000...	723,2644999999...	35,52596774193...	1,542624043059...	240	0,581290150414...
4	F01C004	591674,6330000...	4345037,489000...	723,2174999999...	35,34160804020...	1,541598746132...	199	0,545981427159...
5	F01C005	591673,8660000...	4345038,610999...	723,2464999999...	34,61316129072...	1,658171326036...	155	0,505492419614...
6	F01C006	591673,1409999...	4345039,725999...	723,2345000000...	35,11239999999...	1,577619682328...	175	0,546988372409...
7	F01C007	591672,3560000...	4345040,890000...	723,2285000000...	35,74034285714...	1,352542899713...	175	0,599091703723...
8	F01C008	591671,6099999...	4345041,015999...	723,2105000000...	36,70263513513...	1,740297246284...	148	0,678998855308...
9	F01C009	591670,9070000...	4345042,118999...	723,2184999999...	37,37712707182...	1,405812351310...	181	0,735007544031...
10	F01C010	591670,1479999...	4345043,315999...	723,1894999999...	37,08974358974...	1,399328791934...	195	0,711143680065...
11	F01C011	591669,4340000...	4345045,347000...	723,1964999999...	37,17021608695...	1,462006967822...	184	0,717835134718...
12	F01C012	591668,6050000...	4345046,500000...	723,2545000000...	37,66776470588...	1,331237246711...	170	0,759141610969...
13	F01C013	591667,9459999...	4345047,565999...	723,2454999999...	37,18884651384...	1,525580571259...	156	0,71937301029592...
14	F01C014	591667,1409999...	4345048,786999...	723,2345000000...	38,56699999999...	1,379616097158...	140	0,831812615404...
15	F01C015	591666,3309999...	4345049,912999...	723,2844999999...	37,25627118644...	1,614476858420...	177	0,724971872952...
16	F01C016	591665,6559999...	4345050,950000...	723,2854999999...	37,33823899371...	1,200456946212...	159	0,731778343450...
17	F01C017	591664,9320000...	4345052,126000...	723,2765000000...	37,80440677966...	1,315250932487...	177	0,705718237936...
18	F01C018	591664,1269999...	4345053,305999...	723,2915000000...	36,15977528089...	1,296641448227...	178	0,63392067518134...

Figure 6. Results included in the output shapefile

With this information, thematic maps can be generated (Fig. 7)

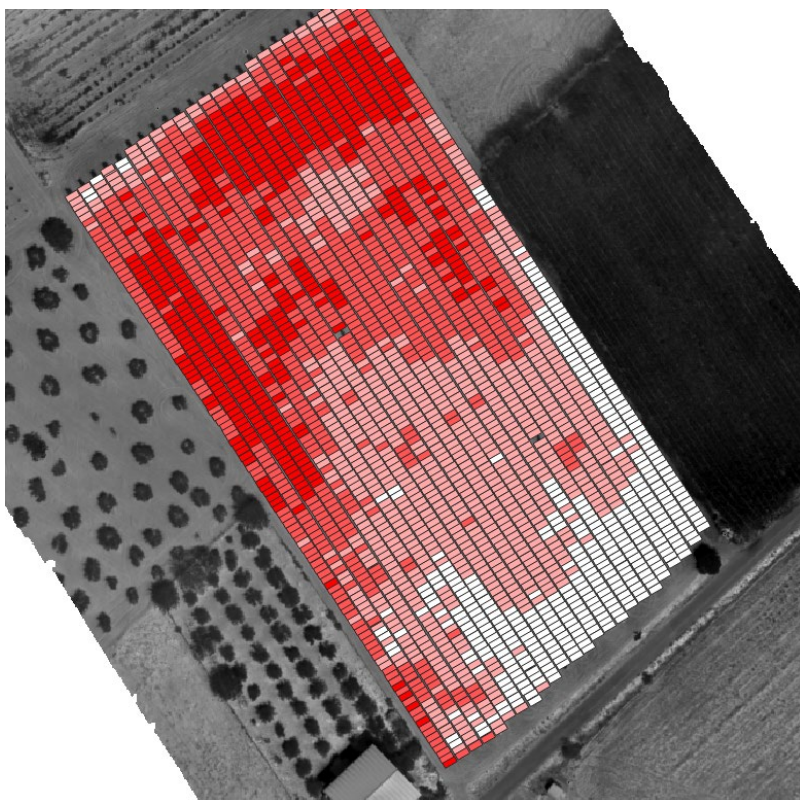


Figure 7. Thematic map of CWSI

To generate alarms for the most stressed vines, the user should select the segmentation method, which can be Kmeans or percentile. If the user selects Kmeans, user should select the number of clusters and the tool will set an alarm for those vines that belongs to the group with higher CWSI values. In the case of using percentiles, the user should specify the percentage of

vines with higher CWSI values, i.e. if the user selects 1%, the tools will set an alarm for the 1% of the vines with higher CWSI values.

List of parameters

Crop Water Stress Index using thermal orthomosaic

	Parameter	Type	Value list	Max	Min	Default value	Description
1	CWSITHO_AirTemperature	Double		60.00	0.00		Air temperature (Celsius degrees) obtained during flight operation.
2	CWSITHO_Date	Text					A valid date should be introduced (i.e. 20220705) or empty if the date is obtained from the thermal orthomosaic.
3	CWSITHO_DateFormat	Text				%%Y%%m%%d	Date format (i.e. 20220705) 5 th July 2022.
4	CWSITHO_DateFromThermal OrthomosaicFile	Value list	True; false				The user decides if obtaining the date from the thermal orthomosaic file name or introduce the date in the previous parameter.

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5	CWSITHO_FactorToTemperature	Double		100000.00	0.000001	0.01	Multiplicative factor to convert raster values in thermal orthomosaic to temperature (Celsius degrees). If the orthomosaic supplies the values directly in temperature, it is 1.
6	CWSITHO_FramesShapeFile	File					The path of the shape file that contains the frame that delimit the region of interest of each vine. The path should not have spaces or special characters.
7	CWSITHO_LowerBaseline CoefficientA	Double		10000.00	-10000.00	-2.016800	Lower base line coefficient A, line slope (adimensional).
8	CWSITHO_LowerBaseline CoefficientB	Double		100.00	-100.00	2.702	Lower base line coefficient B, value of y when x=0 (Celsius degrees).
9	CWSITHO_MaximumValueFor Percentile	Double		1.00	0.01		Percentile value above which the plant's water status is considered anomalous.

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10	CWSITHO_NumberOfClusters ForKmeans	Int		50.00	2.00		Number of groups in which the program will classify the plants based on their CWSI.
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